WORKED SOLUTION

STEP III 2008, Question 8

Pure • Series and Recurrence Relations

STEP III 2008, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q8

The key idea

A linear recurrence for the coefficients means that multiplying the series by the right polynomial makes almost every term cancel. The polynomial is read straight off the recurrence: $a_{r+1} + pa_r = 0$ suggests $1 + px$, and $a_{r+2} + pa_{r+1} + qa_r = 0$ suggests $1 + px + qx^2$. What survives is a short polynomial, so the sum is a rational function. For finite sums, the same trick leaves a few extra terms at the far end. Alternatively, partial fractions split the series into geometric series, which we know how to sum.

Solution

Part (i)

Here the series starts at the x term, so $a_1 = \frac{1}{3}$, $a_2 = \frac{1}{6}$, $a_3 = \frac{1}{12}$. Each coefficient is half the one before, so $a_{r+1} - \frac{1}{2}a_r = 0$ and

$$p = -\tfrac{1}{2}.$$

(There is no constant term. The recurrence links $a_1, a_2, a_3, \ldots$ and says nothing about $r = 0$.) Solving the recurrence gives $a_r = \frac{1}{3}\left(\frac{1}{2}\right)^{r-1}$.

Sum to infinity. Multiply by $1 - \frac{1}{2}x$:

$$\left(1 - \tfrac{1}{2}x\right) S = a_1 x + \sum_{r \geq 1} \left(a_{r+1} - \tfrac{1}{2}a_r\right) x^{r+1} = \tfrac{1}{3}x,$$

because every bracket vanishes by the recurrence. Hence

$$S = \frac{\frac{1}{3}x}{1 - \frac{1}{2}x} = \frac{2x}{3(2 - x)}.$$

This is a geometric series with ratio $\frac{1}{2}x$, so the sum to infinity exists exactly when $|x| < 2$.

The first $n+1$ terms. These are the terms in $x, x^2, \ldots, x^{n+1}$. Write $S_{n+1} = \sum_{r=1}^{n+1} a_r x^r$. Repeating the same multiplication, all the middle terms still cancel, but the top term $-\frac{1}{2}a_{n+1}x^{n+2}$ has no partner:

$$\left(1 - \tfrac{1}{2}x\right) S_{n+1} = \tfrac{1}{3}x - \tfrac{1}{2}a_{n+1}x^{n+2} = \tfrac{1}{3}x - \tfrac{1}{3}\left(\tfrac{1}{2}\right)^{n+1} x^{n+2} = \tfrac{1}{3}x\left(1 - \left(\tfrac{x}{2}\right)^{n+1}\right).$$

So, for $x \neq 2$,

$$S_{n+1} = \frac{2x}{3(2-x)}\left(1 - \left(\frac{x}{2}\right)^{n+1}\right).$$

When $x = 2$ every term equals $\frac{2}{3}$, so $S_{n+1} = \frac{2}{3}(n+1)$.

Part (ii)

Now $a_0 = 2$, $a_1 = 8$, $a_2 = 18$, $a_3 = 37$. Putting $r = 0$ and $r = 1$ into the recurrence gives two equations:

$$18 + 8p + 2q = 0, \qquad 37 + 18p + 8q = 0.$$

The first gives $q = -9 - 4p$. Substituting into the second gives $37 + 18p - 72 - 32p = 0$, so $p = -\frac{5}{2}$ and then $q = 1$. The recurrence is

$$a_{r+2} - \tfrac{5}{2}a_{r+1} + a_r = 0.$$

Sum to infinity. Multiply by $1 + px + qx^2 = 1 - \frac{5}{2}x + x^2$. The coefficient of x^{r+2} in the product is $a_{r+2} - \frac{5}{2}a_{r+1} + a_r = 0$ for every $r \geq 0$. Only the x^0 and x^1 coefficients survive:

$$\left(1 - \tfrac{5}{2}x + x^2\right) T = a_0 + \left(a_1 - \tfrac{5}{2}a_0\right) x = 2 + 3x.$$

Doubling top and bottom, and factorising $2 - 5x + 2x^2 = (1-2x)(2-x)$,

$$T = \frac{4 + 6x}{(1-2x)(2-x)}.$$

Partial fractions. Write $\dfrac{4+6x}{(1-2x)(2-x)} = \dfrac{A}{1-2x} + \dfrac{B}{2-x}$. Cover-up at $x = \frac{1}{2}$ gives $A = \dfrac{7}{3/2} = \frac{14}{3}$, and at $x = 2$ gives $B = \dfrac{16}{-3} = -\frac{16}{3}$. So

$$T = \frac{14}{3} \cdot \frac{1}{1-2x} - \frac{8}{3} \cdot \frac{1}{1 - \frac{1}{2}x}.$$

Expanding each term as a geometric series,

$$a_r = \tfrac{14}{3}\, 2^r - \tfrac{8}{3}\left(\tfrac{1}{2}\right)^r.$$

Check: $r = 0, 1, 2, 3$ give 2, $\frac{28}{3} - \frac{4}{3} = 8$, $\frac{56}{3} - \frac{2}{3} = 18$ and $\frac{112}{3} - \frac{1}{3} = 37$, as they should.

Ask yourself For which x does the sum to infinity actually exist?

We need both geometric series to converge, so $|2x| < 1$ and $|\frac{1}{2}x| < 1$. The first condition is the stricter one, so we need $|x| < \frac{1}{2}$.

The first $n+1$ terms. These are the terms in x^0 to x^n. Summing each geometric series to $n+1$ terms gives, for $x \neq \frac{1}{2}, 2$,

$$\sum_{r=0}^{n} a_r x^r = \frac{14}{3} \cdot \frac{1-(2x)^{n+1}}{1-2x} - \frac{16}{3} \cdot \frac{1-\left(\frac{x}{2}\right)^{n+1}}{2-x}.$$

(The second fraction is $\frac{8}{3} \cdot \frac{1-(x/2)^{n+1}}{1-x/2}$, with top and bottom multiplied by 2.) If $x = \frac{1}{2}$ or $x = 2$, the matching geometric sum is just $n+1$ copies of 1.

Checks

- Let $n \to \infty$ with $|x| < \frac{1}{2}$: the powers $(2x)^{n+1}$ and $(x/2)^{n+1}$ vanish and we recover the partial-fraction form of T. The same happens in part (i) with $|x| < 2$.
- With $n = 0$ the part (i) formula gives $\frac{2x}{3(2-x)} \cdot \frac{2-x}{2} = \frac{1}{3}x$. The part (ii) formula gives $\frac{14}{3} - \frac{16}{3} \cdot \frac{1}{2} = 2$. Both are the first term, as they should be.
- Numerically, at $x = 0.3$ we have $T = \dfrac{5.8}{0.4 \times 1.7} \approx 8.5294$, and the partial sums of $2 + 8x + 18x^2 + 37x^3 + \cdots$ approach this value.

Common slips

- In part (i), taking the first term to be $a_0 x^0$. The series starts at x^1, so the first $n+1$ terms run up to x^{n+1}.
- Forgetting the leftover end terms when multiplying a *finite* sum by $1 + px$. These terms are exactly what produce the $(\frac{x}{2})^{n+1}$ factor.
- Quoting $|x| < 2$ for T. The root nearer the origin, $x = \frac{1}{2}$, sets the radius of convergence.

Beyond STEP

The power series $\sum a_r x^r$ is the *generating function* of the sequence (a_r). A linear recurrence with constant coefficients always gives a rational generating function, and the poles of that function decide both the closed form for a_r and the radius of convergence. Generating functions are a standard tool in combinatorics and probability, and the radius of convergence of power series is developed carefully in first-year analysis. Abbott's *Understanding Analysis* covers the convergence side well.

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